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ANALYSIS OF DISCRETE ILL-POSED PROBLEMS BY MEANS OF THE L-CURVE*

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Abstract. When discrete ill-posed problems are analyzed and solved by various numerical regularization techniques, a very convenient way to display information about the regularized solution is to plot the norm or seminorm of the solution versus the norm of the residual vector. In particular, the graph associated with Tikhonov regularization plays a central role. The main purpose of this paper is to advocate the use of this graph in the numerical treatment of discrete ill-posed problems. The graph is characterized quantitatively, and several important relations between regularized solutions and the graph are derived. It is also demonstrated that several methods for choosing the regularization parameter are related to locating a characteristic L-shaped "corner" of the graph.

Key words. discrete ill-posed problems, least squares, generalized SVD, regularization

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1. Introduction.

We say that the algebraic problems $A\mathbf{x} = \mathbf{b}$ and $\min \|A\mathbf{x} - \mathbf{b}\|$ are *discrete ill-posed problems* if the matrix A is ill conditioned and all its singular values decay to zero in such a way that there is no particular gap in the singular value spectrum. Discrete ill-posed problems arise in a variety of applications: astronomy [5], computerized tomography [32], early vision [2], electrocardiography [7], mathematical physics [41], and meteorology [46], to mention just a few. The underlying mathematical problem is often—but not always—a linear Fredholm integral equation of the first kind.

There is a vast amount of literature on ill-posed problems in the setting of Hilbert-spaces and other infinite-dimensional spaces; see, e.g., [10], [11], [13], [14], [27], [31], [40], [47]. The approach taken in this paper is different: we take the algebraic problems $A\mathbf{x} = \mathbf{b}$ and $\min \|A\mathbf{x} - \mathbf{b}\|$ as our basis, and we use numerical linear algebra—in particular, the generalized singular value decomposition—to derive our results. Introductions to discrete ill-posed problems can be found in [4], [5], [29], [33], [35], [43], [44]. The monograph by Hofmann [25] contains a wealth of material on infinite-dimensional as well as finite-dimensional ill-posed problems.

Most numerical methods for treating discrete ill-posed problems seek to overcome the problems associated with the large condition number of A by replacing the problem with a "nearby" well-conditioned problem whose solution approximates the required solution and, in addition, is a more satisfactory solution than the ordinary (least squares) solution. The latter goal is achieved by incorporating additional information about the sought solution, often that the computed solution should be smooth. Such methods are called *regularization methods*, and they always include a so-called regularization parameter λ which controls the degree of smoothing or regularization applied to the problem. As the regularization parameter λ varies, we obtain regularized solutions $\mathbf{x}_\lambda$ having properties that vary with λ. A convenient way to display and understand these properties is to plot the norm—or, more generally, a seminorm—of the regularized solution, $\|L\mathbf{x}_\lambda\|$, versus the norm of the corresponding residual vector, $\|A\mathbf{x}_\lambda - \mathbf{b}\|$. This was originally suggested in the classic book by Lawson and Hanson [28].

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In this paper, our aim is to show how such plots reveal considerable information about the discrete ill-posed problem as well as about the particular regularization method used. We will also demonstrate how these plots are a valuable aid in choosing a good (i.e., nearly optimal) regularization parameter.

We stress that although the mere restriction of an infinite-dimensional problem to a finite-dimensional one (e.g., by discretization of the problem) exhibits some regularizing effect, it usually does not provide enough regularization for practical purposes; cf. [6, §5].

For continuous problems, the choice of norms for measuring the solution as well as the residual plays a central role. However, for discrete problems the standard norms are equivalent; for example, if $\mathbf{x} \in \mathbb{R}^n$ and $A \in \mathbb{R}^{m \times n}$, then $\|\mathbf{x}\|_\infty \leq \|\mathbf{x}\|_2 \leq \sqrt{n}\,\|\mathbf{x}\|_\infty$ and $\|A\|_2 \leq \|A\|_F \leq \sqrt{\min(m,n)}\,\|A\|_2$, where $\|\mathbf{x}\|_\infty \equiv \max_i |x_i|$ and $\|A\|_F \equiv \left(\sum_{i=1}^m \sum_{j=1}^n a_{ij}^2\right)^{1/2}$. The generalized singular value decomposition—the superior "tool" for analysis of discrete regularized problems—is intimately connected to 2-norms. Throughout the paper we, therefore, deal entirely with vector and matrix 2-norms, which we denote by $\|\cdot\|$. The 2-norm is a natural choice for measuring the residual vector as long as outliers are of no concern. We also feel that the seminorm $\|L\,\mathbf{x}\|$ of the solution (with the norm $\|\mathbf{x}\|$ as a special case) is appropriate for many problems.

The paper is organized as follows. In §2 we give a brief introduction to discrete regularization methods. Section 3 introduces the L-curve and gives a description of its characteristic L-shaped appearance. Sections 4 and 5 treat different aspects of similarities between Tikhonov regularization and other regularization methods. In §6 we show how different methods for choosing the regularization parameter are related to finding a regularized solution near the L-shaped "corner" of the L-curve. Finally, in §7 we illustrate these topics by numerical examples.

2. Numerical regularization methods. As mentioned above, when discrete ill-posed problems $A\,\mathbf{x} = \mathbf{b}$ and $\min \|A\,\mathbf{x} - \mathbf{b}\|$ are solved numerically, some sort of regularization is needed to ensure that the computed, regularized solution $\mathbf{x}_\lambda$ is not too sensitive to perturbations of A and $\mathbf{b}$ and, in addition, has a suitably small seminorm $\|L\,\mathbf{x}_\lambda\|$. Here, L is a matrix with full row rank, typically a discrete approximation to some derivative operator. The rationale behind the latter goal is that the solution to a physical problem usually has a small norm or seminorm. Both goals are achieved at the same time by imposing the regularization on the solution.

To see how this is achieved, let us introduce the *generalized singular value decomposition* (GSVD) of the matrix pair (A, L). For the problems that we are considering, with $A \in \mathbb{R}^{m \times n}$, $L \in \mathbb{R}^{p \times n}$, and $m \geq n \geq p$, the GSVD can be written as follows:

$$A = U\,\Sigma\,X^{-1}, \qquad L = V\,M\,X^{-1}. \tag{1}$$

Here, $U \in \mathbb{R}^{m \times n}$ and $V \in \mathbb{R}^{p \times p}$ have orthonormal columns such that $U^T U = I_n$ and $V^T V = I_p$; $X \in \mathbb{R}^{n \times n}$ is a nonsingular matrix, and Σ and M are of the form

$$\Sigma = \begin{pmatrix} \Sigma_p & 0 \\ 0 & I_{n-p} \end{pmatrix}, \qquad M = \begin{pmatrix} M_p & 0 \end{pmatrix}. \tag{2}$$

The matrices $\Sigma_p = \operatorname{diag}(\sigma_i)$ and $M_p = \operatorname{diag}(\mu_i)$ are both $p \times p$ diagonal matrices whose diagonal elements satisfy $\sigma_i^2 + \mu_i^2 = 1$ and are ordered such that

$$0 \leq \sigma_1 \leq \cdots \leq \sigma_p, \qquad 1 \geq \mu_1 \geq \cdots \geq \mu_p > 0. \tag{3}$$

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For a proof of this decomposition, see, e.g., [3, §22]. The *generalized singular values* γ_i of (A, L) are defined as the positive quantities

$$\gamma_i \equiv \frac{\sigma_i}{\mu_i}, \qquad i = 1, \ldots, p. \tag{4}$$

In particular, if $L = I_n$ then $X^{-1} = M^{-1} V$ and $A = U\,(\Sigma\, M^{-1})\, V^T$, showing that the generalized singular values in this special case are related to the ordinary singular values ψ_i of A by $\psi_i = \gamma_{n-i+1}$, $i = 1, \ldots, n$. As we shall see below, regularized solutions can be expressed in a very convenient way in terms of the GSVD of (A, L).

One of the best known regularization methods is *Tikhonov regularization* [40] (also called damped least squares [3, §26]). The Tikhonov regularized solution $\mathbf{x}_\lambda$ is defined as the solution to the following least squares problem:

$$\min_{\mathbf{x}} \{\, \|A\,\mathbf{x} - \mathbf{b}\|^2 + \lambda^2 \|L\,\mathbf{x}\|^2 \,\}. \tag{5}$$

Here, λ controls the weight given to minimization of the seminorm $\|L\,\mathbf{x}\|$ of the solution relative to minimization of the residual norm $\|A\,\mathbf{x} - \mathbf{b}\|$. It is straightforward to show that the solution to (5) is given by

$$\mathbf{x}_\lambda = \sum_{i=1}^{p} \frac{\gamma_i^2}{\gamma_i^2 + \lambda^2} \frac{\mathbf{u}_i^T \mathbf{b}}{\sigma_i}\, \mathbf{x}_i + \sum_{i=p+1}^{n} (\mathbf{u}_i^T \mathbf{b})\, \mathbf{x}_i. \tag{6}$$

Notice how the *filter factors* $\gamma_i^2/(\gamma_i^2 + \lambda^2)$ for Tikhonov regularization in effect dampen, or filter out, the contributions to $\mathbf{x}_\lambda$ corresponding to the generalized singular values γ_i smaller than about λ. Since $\sigma_i = \gamma_i(1 - \sigma_i^2)^{1/2} \approx \gamma_i$ for all $\sigma_i \ll 1$, and since the largest perturbations of the ordinary least squares solution are associated with the smallest σ_i, it is clear that the regularized solution $\mathbf{x}_\lambda$ will be less sensitive to perturbations than the ordinary least squares solution. In fact, it is shown in [19] that the condition number for the problem (5) is $\|A\|\,\|X\|/\lambda$. In addition, it can be shown that the number of oscillations in $\mathbf{x}_i$ (i.e., the number of sign changes in the elements of $\mathbf{x}_i$) increases as γ_i decreases; i.e., the smaller the γ_i, the more oscillatory the $\mathbf{x}_i$, such that $\mathbf{x}_\lambda$ is indeed smoother than the unregularized solution [18].

An interesting aspect of many numerical regularization techniques is that they, from a practical point of view, produce the same regularized solutions. By this, we mean that both the regularized solutions and the corresponding residual vectors are practically identical, provided of course that reasonable regularization parameters are used for each method. One manifestation of this fact is that these regularized solutions have approximately the same expansion in terms of the GSVD of (A, L). For example, the truncated GSVD method [18], [21] (which is identical to the classical truncated SVD method when $L = I_n$) leads to a regularized solution given by

$$\mathbf{x}_k = \sum_{i=p-k+1}^{p} \frac{\mathbf{u}_i^T \mathbf{b}}{\sigma_i}\, \mathbf{x}_i + \sum_{i=p+1}^{n} (\mathbf{u}_i^T \mathbf{b})\, \mathbf{x}_i, \tag{7}$$

corresponding to filter factors zero and 1. If the Lanczos bidiagonalization process [3, §20] is halted after q steps, and a least squares solution is computed on the basis of the $(q+1) \times q$ bidiagonal matrix [34], then it can be shown that the associated L is the identity matrix while the associated filter factors are $1 - R_q(\psi_i^2)$, where R_q denotes the qth degree Ritz polynomial. The analysis in [24], [42, Thm. 6.7] and [45] shows that Lanczos

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bidiagonalization is similar to Tikhonov regularization with $L = I_n$. An analogous result holds for Richardson/Landweber/Fridman/Picard/Cimino iterations [9], [46]. Several other examples are given in [9, §4]. As a consequence, we can limit our discussion in this paper to the Tikhonov regularization method, knowing that our results carry over to these regularization methods as well.

Examples of schemes that do not have simple expressions for the filter factors are the maximum entropy principle [37] and the regularization method proposed by Babolian and Delves [1]:

$$\min \|A\,\mathbf{x} - \mathbf{b}\| \quad \text{subject to } |x_j| \le C\,j^{-r}, \quad j = 1, \ldots, n, \tag{8}$$

where the components x_j of $\mathbf{x}$ are coefficients in a Chebychev expansion of a continuous solution, and where C and r are constants.

3. The L-curve and its properties. A convenient way to display information about the regularized solution $\mathbf{x}_\lambda$ to (5), as a function of the regularization parameter λ, is to plot the norm or seminorm $\|L\,\mathbf{x}_\lambda\|$ of the solution versus the residual norm $\|A\,\mathbf{x}_\lambda - \mathbf{b}\|$. In this way, one can easily get an idea of the compromise between the minimization of these two quantities. One can also immediately see whether one or both quantities are unreasonably large (or small). The same is true for other regularization methods as well, and the technique—perhaps combined with other criteria—serves as a practical guide to choosing a good regularization parameter. We return to this subject in §6.

The practical use of such plots was first suggested by Lawson and Hanson [28, Chap. 25 and 26]. Similar plots also appear in [30] and, more recently, in [39], [17], [20], and [23]. In fact, it seems that a number of researchers have used these plots in the practical treatment of ill-posed problems—although they have rarely made their way to the final publication.

As mentioned above, we restrict this presentation to Tikhonov regularization, and we define the associated *L-curve* as the continuous curve consisting of all the points $(\|A\,\mathbf{x}_\lambda - \mathbf{b}\|, \|L\,\mathbf{x}_\lambda\|)$ for $\lambda \in [0, \infty)$. A number of important properties of this curve are summarized below, but first we need the following expression for the unregularized solution, defined as the limit of $\mathbf{x}_\lambda$ for $\lambda \to 0$:

$$\mathbf{x}_0 \equiv \lim_{\lambda \to 0} \mathbf{x}_\lambda = X\,\Sigma^+\,U^T\mathbf{b}. \tag{9}$$

We also need the extreme residual norms corresponding to zero and infinite regularization, respectively:

$$\delta_0 \equiv \|(I_m - U\,U^T)\mathbf{b}\|, \quad \delta_\infty \equiv \delta_0 + \|U_p\,U_p^T\mathbf{b}\|, \quad U_p = [\mathbf{u}_1, \ldots, \mathbf{u}_p]. \tag{10}$$

Here, δ_0 is the norm of that component of the right-hand side $\mathbf{b}$ which is outside the range of A. The system is consistent if $\delta_0 = 0$, and δ_0 is, therefore, sometimes called the *incompatibility measure*. With this notation, the L-curve has the following properties.

THEOREM 1. *Let $\mathbf{x}_\lambda$ denote the solution to* (5). *Then $\|L\,\mathbf{x}_\lambda\|$ is a monotonically decreasing function of $\|A\,\mathbf{x}_\lambda - \mathbf{b}\|$, and any point (δ, η) on the curve $(\|A\,\mathbf{x}_\lambda - \mathbf{b}\|, \|L\,\mathbf{x}_\lambda\|)$ is a solution to the following two inequality-constrained least squares problems*:

$$\delta = \min \|A\,\mathbf{x} - \mathbf{b}\| \text{ subject to } \|L\,\mathbf{x}\| \le \eta, \qquad 0 \le \eta \le \|L\,\mathbf{x}_0\|, \tag{11}$$

$$\eta = \min \|L\,\mathbf{x}\| \text{ subject to } \|A\,\mathbf{x} - \mathbf{b}\| \le \delta, \qquad \delta_0 \le \delta \le \delta_\infty. \tag{12}$$

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Proof. The fact that $\|L\,\mathbf{x}_\lambda\|$ is a decreasing function of $\|A\,\mathbf{x}_\lambda - \mathbf{b}\|$ follows immediately from the following expressions:

$$\|L\,\mathbf{x}_\lambda\|^2 = \sum_{i=1}^{p} \left(\frac{\gamma_i^2}{\gamma_i^2 + \lambda^2} \frac{\mathbf{u}_i^T \mathbf{b}}{\sigma_i} \right)^2, \tag{13}$$

$$\|A\,\mathbf{x}_\lambda - \mathbf{b}\|^2 = \sum_{i=1}^{p} \left(\frac{\lambda^2}{\gamma_i^2 + \lambda^2} \mathbf{u}_i^T \mathbf{b} \right)^2 + \delta_0^2, \tag{14}$$

which are easily derived by means of (6). The second part of the theorem is a standard result in the theory of Tikhonov regularization; cf., e.g., [28, Thm. (25.49)]. □

According to Theorem 1, the L-curve $(\|A\,\mathbf{x}_\lambda - \mathbf{b}\|, \|L\,\mathbf{x}_\lambda\|)$ divides the first quadrant into two separate regions. It is impossible to construct any solution that corresponds to a point below the L-curve, and any regularized solution $\mathbf{x}_{\text{reg}}$ must necessarily lie on or above the L-curve. The Tikhonov regularized solution $\mathbf{x}_\lambda$ is optimal in the sense that, given η (or δ), there does not exist a solution with a smaller residual norm (or solution seminorm) than $\|A\,\mathbf{x}_\lambda - \mathbf{b}\|$ (or $\|L\,\mathbf{x}\|$). A convenient way to characterize the quality of other regularized solutions $\mathbf{x}_{\text{reg}}$ is, therefore, to measure the deviations $L\,\mathbf{x}_\lambda - L\,\mathbf{x}_{\text{reg}}$ and $(A\,\mathbf{x}_\lambda - \mathbf{b}) - (A\,\mathbf{x}_{\text{reg}} - \mathbf{b})$ by giving upper bounds for the norm of these quantities. The smaller the bounds, the closer $(\|A\,\mathbf{x}_{\text{reg}} - \mathbf{b}\|, \|L\,\mathbf{x}_{\text{reg}}\|)$ is to the L-curve. Examples of such investigations for the truncated SVD and the truncated GSVD methods can be found in [15], [20], [21]. More details are given in §5.

If a few reasonable assumptions are made about the ill-posed problem, then it is possible to characterize the L-curve more completely than in Theorem 1. This was first done in [20] for the case $L = I_n$. Here, we generalize the analysis to general matrices L with full row rank as they appear in discrete regularization methods.

CHARACTERIZATION 2. *Let* $\bar{\mathbf{b}}$ *denote the unperturbed right-hand side, and let* $\mathbf{e}$ *denote the perturbation (i.e., the errors) in* $\mathbf{b}$ *such that* $\mathbf{b} = \bar{\mathbf{b}} + \mathbf{e}$. *Assume that*

1. *The coefficients* $|\mathbf{u}_i^T \bar{\mathbf{b}}|$ *on average decay to zero faster than the* γ_i,
2. *The perturbation vector* $\mathbf{e}$ *has zero mean and covariance matrix* $\sigma_0^2 I_m$, *and*
3. *The norm of* $\mathbf{e}$ *satisfies* $\|\mathbf{e}\| < \|\bar{\mathbf{b}}\|$.

Then the L-curve $(\|A\,\mathbf{x}_\lambda - \mathbf{b}\|, \|L\,\mathbf{x}_\lambda\|)$ *exhibits a "corner" behavior as a function of* λ, *and the "corner" appears approximately at* $(\sqrt{\sigma_0^2(m-n+p) + \delta_0^2}\,, \|L\,\bar{\mathbf{x}}_0\|)$. *Here,* $\bar{\mathbf{x}}_0$ *is the unregularized solution* (9) *to the unperturbed problem, and* δ_0 *is the incompatibility measure* (10). *The larger the difference between the decay rates of* $|\mathbf{u}_i^T \bar{\mathbf{b}}|$ *and* $|\mathbf{u}_i^T \mathbf{e}|$, *the more distinct the "corner" will appear.*

We cannot give a detailed and rigorous proof of this characterization, but instead we summarize the reasoning that leads to it. Assumption 1 is in fact the discrete Picard condition [21], and it is necessary in order to ensure that a reasonable regularized solution exists at all. Assumption 2 ensures that the solution $\mathbf{x}_\lambda$ has a reasonable covariance matrix (if it is not satisfied, we should rescale the problem or, if necessary, use the general Gauss–Markov linear model [48] for general covariance matrices). Finally, assumption 3 ensures a reasonable signal-to-noise ratio in the given right-hand side $\mathbf{b}$.

Let us first consider the behavior of the L-curve for an unperturbed problem with $\mathbf{e} = \mathbf{0}$. For $\lambda \ll \gamma_n$, we have $\gamma_i^2 + \lambda^2 \approx \gamma_i^2$ such that $\gamma_i^2/(\gamma_i^2 + \lambda^2) \approx 1$, $\mathbf{x}_\lambda \approx \bar{\mathbf{x}}_0$, and

$$\|A\,\mathbf{x}_\lambda - \bar{\mathbf{b}}\|^2 \approx \sum_{i=1}^{p} \left(\frac{\lambda^2}{\gamma_i^2} \mathbf{u}_i^T \bar{\mathbf{b}} \right)^2 + \delta_0^2 \le \left(\left(\frac{\lambda}{\gamma_n} \right)^2 \|\bar{\mathbf{b}}\| \right)^2 + \delta_0^2.$$

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Hence, for small λ the L-curve is approximately a horizontal line at $\|L\,\mathbf{x}_\lambda\| = \|L\,\bar{\mathbf{x}}_0\|$. As λ increases, if follows from (13) that $\|L\,\mathbf{x}_\lambda\|$ starts to decrease, while $\|A\,\mathbf{x}_\lambda - \mathbf{b}\|$ still grows towards δ_∞. The L-curve eventually must start to bend down towards the abscissa axis, which happens when λ is comparable with the largest generalized singular values γ_i. For those values of λ, the residual norm is still somewhat smaller than δ_∞ because some of the coefficients $\lambda^2/(\gamma_i^2+\lambda^2)$ in the expression (14) for $\|A\,\mathbf{x}_\lambda - \mathbf{b}\|$ are less than one.

Consider now the L-curve associated with the mere perturbation $\mathbf{e}$ of the right-hand side. The corresponding "solution" $\mathbf{x}_\lambda^{(e)}$, given by (6) with $\mathbf{b}$ replaced by $\mathbf{e}$, satisfies (from assumption 2)

$$\|L\,\mathbf{x}_\lambda^{(e)}\|^2 = \sum_{i=1}^{p} \left(\frac{\gamma_i}{\gamma_i^2+\lambda^2}\,\mathbf{u}_i^T\mathbf{e} \right)^2 \approx \sigma_0^2 \left(\sum_{i=1}^{p} \frac{\gamma_i}{\gamma_i^2+\lambda^2} \right)^2,$$

$$\begin{aligned} \|A\,\mathbf{x}_\lambda^{(e)} - \mathbf{b}\|^2 &= \sum_{i=1}^{p} \left(\frac{\lambda^2}{\gamma_i^2+\lambda^2}\,\mathbf{u}_i^T\mathbf{e} \right)^2 + \|(I_m - U\,U^T)\,\mathbf{e}\|^2 \\ &\approx \sigma_0^2 \left(\sum_{i=1}^{p} \left(\frac{\lambda^2}{\gamma_i^2+\lambda^2} \right)^2 + (m-n) \right). \end{aligned}$$

For small $\lambda \ll \gamma_n$, this L-curve is approximately a horizontal line at $\|L\,\mathbf{x}_\lambda^{(e)}\| \approx \sqrt{p}\,\sigma_0/\gamma_n$, and it starts to bend down towards the abscissa axis for much smaller values of λ than the L-curve for $\bar{\mathbf{b}}$, namely, when λ is of the same size as the smallest generalized singular values. Moreover, we see that as λ increases, $\|A\,\mathbf{x}_\lambda^{(e)} - \mathbf{b}\|$ becomes almost independent of λ, while $\|L\,\mathbf{x}_\lambda^{(e)}\|$ is dominated by a few terms (ℓ, say) for which $\gamma_i/(\gamma_i^2+\lambda^2) \approx 1/(2\lambda)$ such that $\|L\,\mathbf{x}_\lambda^{(e)}\| \approx \ell\sigma_0/(2\lambda) \approx \sigma_0/\lambda$. Hence, this L-curve soon becomes almost a vertical line at $\|A\,\mathbf{x}_\lambda - \mathbf{b}\| \approx \sigma_0\sqrt{m-n+p}$ as $\lambda \to \infty$.

The actual L-curve for a given problem, with a perturbed right-hand side $\mathbf{b} = \bar{\mathbf{b}} + \mathbf{e}$, is a combination of the above two special L-curves. For small λ the behavior of the L-curve is entirely dominated by contributions from $\mathbf{e}$, while for large λ it is completely dominated by contributions from $\bar{\mathbf{b}}$. In between, there is a small region where both $\bar{\mathbf{b}}$ and $\mathbf{e}$ contribute, and this region defines the L-shaped "corner" of the L-curve. Moreover, the faster the coefficients $|\mathbf{u}_i^T\bar{\mathbf{b}}|$ decay to zero, the smaller this cross-over region and, thus, the sharper the L-shaped "corner." This explains our choice of the name "L-curve." More properties of the L-curve are derived by Hansen and O'Leary in [23] where it is also shown that the characteristic L-shaped corner is most pronounced in a log-log plot. For examples of L-curves, see Figs. 1 and 4 in §7.

It seems intuitively clear that a good regularization parameter λ is one that corresponds to a regularized solution near the "corner" of the L-curve because in this region there is a good compromise between achieving a small residual norm $\|A\,\mathbf{x}_\lambda - \mathbf{b}\|$ and keeping the solution seminorm $\|L\,\mathbf{x}_\lambda\|$ reasonably small. As we shall see in §6, this is indeed the case.

4. The similarity of regularized solutions. As already mentioned in §3, a convenient way to characterize a regularized solution $\mathbf{x}_{\text{reg}}$ is to measure how far it is from the L-curve associated with Tikhonov regularization. The closer $(\|A\,\mathbf{x}_{\text{reg}} - \mathbf{b}\|, \|L\,\mathbf{x}_{\text{reg}}\|)$ is to the L-curve, the better $\mathbf{x}_{\text{reg}}$ is in the "Tikhonov sense." Therefore, one can readily think of a narrow band lying above the L-curve defining an area in which any solution is a satisfactory regularized solution. It is interesting to give a more quantitative characterization of solutions within this band-shaped area. In particular, if we are given two

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solutions, $\mathbf{x}_1$ and $\mathbf{x}_2$, both of which have seminorms less than η and residual norms less than δ, what can be said about $\mathbf{x}_1 - \mathbf{x}_2$? Obviously, we have $\|L(\mathbf{x}_1 - \mathbf{x}_2)\| \le 2\eta$, but we can improve on this result by means of the following theorem.

THEOREM 3. *Given two solutions, $\mathbf{x}_1$ and $\mathbf{x}_2$, both satisfying*

$$\|L\mathbf{x}_i\| \le \eta, \quad \|A\mathbf{x}_i - \mathbf{b}\| \le \delta, \quad i = 1, 2, \tag{15}$$

then their difference $\mathbf{x}_1 - \mathbf{x}_2$ satisfies

$$\|\mathbf{x}_1 - \mathbf{x}_2\| \le 2\,\|X\|\sqrt{\delta^2 + \eta^2}, \tag{16}$$

$$\|L(\mathbf{x}_1 - \mathbf{x}_2)\| \le 2\min\left\{\frac{\delta}{\gamma_1}, \eta\right\}, \tag{17}$$

$$\|A(\mathbf{x}_1 - \mathbf{x}_2)\| \le 2\min\{\delta, \eta\gamma_p\}. \tag{18}$$

Proof. We first make a change of variable to $\xi = X^{-1}\mathbf{x}$, such that $\|L\mathbf{x}\| = \|M\xi\|$, $\|A\mathbf{x}\| = \|\Sigma\xi\|$, and $\|\mathbf{x}\| \le \|X\|\,\|\xi\|$. Following Miller [30] we then need to find the so-called stability estimate (where the matrix C is M, Σ, or I_n):

$$\mathcal{M}(\delta, \eta, C) \equiv \sup\{\,\|C\xi\| : \|M\xi\| \le \eta,\ \|\Sigma\xi\| \le \delta\,\},$$

for then $\|C(\xi_1 - \xi_2)\| \le 2\mathcal{M}(\delta, \eta, C)$. Using the GSVD of (A, L), we can easily see that $\|M\xi\|$ is bounded either by η (whenever the constraint $\|M\xi\| = \eta$ is active) or by $\max\{\delta/\gamma_1, \ldots, \delta/\gamma_p\} = \delta/\gamma_1$ (when the constraint $\|\Sigma\xi\| = \delta$ is active), so that $\mathcal{M}(\delta, \eta, M) = \min\{\delta/\gamma_1, \eta\}$. By a similar argument, we get $\mathcal{M}(\delta, \eta, \Sigma) = \min\{\delta, \eta\gamma_p\}$. This yields (17) and (18). To find $\mathcal{M}(\delta, \eta, I_n)$, we use the relation $\sigma_i^2 + \mu_i^2 = 1$ to obtain

$$\mathcal{M}(\delta, \eta, I_n) \le \max_{0 \le \sigma \le 1}\left\{\min\left\{\frac{\delta}{\sigma}, \frac{\eta}{\sqrt{1 - \sigma^2}}\right\}\right\}.$$

Solving for σ yields $\sigma = \delta(\eta^2 + \delta^2)^{-1/2} \Rightarrow \delta/\sigma = \sqrt{\delta^2 + \eta^2}$, and we obtain (16). □

Although the upper bounds in Theorem 3 are attained in very special cases only, these results are still interesting, partly because they provide information about $\mathbf{x}_1 - \mathbf{x}_2$ (and not just $L(\mathbf{x}_1 - \mathbf{x}_2)$ and $A(\mathbf{x}_1 - \mathbf{x}_2)$), and partly because they combine the tolerances δ and η. For example, we see from (18) that a small tolerance η for the solution seminorm ensures that $\mathbf{x}_1$ and $\mathbf{x}_2$ have very similar residual vectors, while (17) tells us that a small residual tolerance δ does not ensure that $L\mathbf{x}_1$ and $L\mathbf{x}_2$ are close. Concerning the bound (16) for $\mathbf{x}_1 - \mathbf{x}_2$, it should be noted that $\|X\| \approx \|L^+\|$ (where L^+ is the pseudo-inverse of L) and that L usually is a fairly well-conditioned matrix; cf. [22].

5. The similarity with Tikhonov regularization. Another interesting question, which is related to the question discussed above, is the following: given a solution $\mathbf{x}_{\text{reg}}$ computed by a regularization method different from Tikhonov's method, is this $\mathbf{x}_{\text{reg}}$ similar to the solution $\mathbf{x}_\lambda$ computed by means of Tikhonov regularization for some λ? Or, formulated in terms of the L-curve, is the point $(\|A\mathbf{x}_{\text{reg}} - \mathbf{b}\|, \|L\mathbf{x}_{\text{reg}}\|)$ close to the L-curve for Tikhonov regularization? It turns out that without any extra information about the right-hand side $\mathbf{b}$ we cannot answer this question. To see why this is so, we recall that by means of the GSVD we can write $L\mathbf{x}_\lambda$ and $L\mathbf{x}_{\text{reg}}$ in the convenient form

$$L\mathbf{x}_\lambda = V\Phi M_p\Sigma_p^{-1}U_p^T\mathbf{b}, \qquad L\mathbf{x}_{\text{reg}} = VFM_p\Sigma_p^{-1}U_p^T\mathbf{b},$$

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where Φ and F are $p \times p$ diagonal matrices with diagonal elements equal to the filter factors for the two regularization methods, namely, $\phi_i = \gamma_i^2/(\gamma_i^2 + \lambda^2)$ and f_i. An upper bound for the norm of the difference between $L\,\mathbf{x}_\lambda$ and $L\,\mathbf{x}_{\rm reg}$ that only involves the norm of the right-hand side $\mathbf{b}$ is then given by

$$\|L\,(\mathbf{x}_\lambda - \mathbf{x}_{\rm reg})\| \le \|(\Phi - F)M_p\Sigma_p^{-1}\|\,\|\mathbf{b}\| = \max\left\{\frac{|\phi_i - f_i|}{\gamma_i}\right\}\|\mathbf{b}\|. \tag{19}$$

Unfortunately, the quantity $\max\{|\phi_i - f_i|/\gamma_i\}$ can easily be of the same order as $\|L\,\mathbf{x}_\lambda\|$ or $\|L\,\mathbf{x}_{\rm reg}\|$, even in the case where Φ and F produce similar regularized solutions.

As an example of this, let $\mathbf{x}_{\rm reg}$ be the truncated GSVD solution $\mathbf{x}_k$ given by (7). If the integer k satisfies $\gamma_{p-k} \le \lambda \le \gamma_{p-k+1}$, then it is proved in [21, Thm. 3][1] that

$$\min\left\{\max\left\{\frac{|\phi_i - f_i|}{\gamma_i}\right\}\right\} \ge \frac{1}{2}\left(\frac{\gamma_{p-k}}{\gamma_{p-k+1}}\right)^{1/2}\|L\,\mathbf{x}_{\rm reg}\|$$

and yet $L\,\mathbf{x}_\lambda$ and $L\,\mathbf{x}_{\rm reg}$ can be very close *independently* of the γ_i-spectrum [21, Thm. 4]. A numerical example of this is shown in Fig. 1 in §7. Since $0 \le \gamma_{p-k}/\gamma_{p-k+1} \le 1$, this example illustrates that the right-hand side in (19) can indeed be of the order $\|L\,\mathbf{x}_{\rm reg}\|\,\|\mathbf{b}\|$ even if $L\,\mathbf{x}_\lambda$ and $L\,\mathbf{x}_{\rm reg}$ are very close.

A simple alternative to the upper bound in (19) is $\|\Phi - F\|\,\|M_p\Sigma_p^{-1}U_p^T\mathbf{b}\|$, but this is also inadequate for our purpose because it does not take into account the markedly different behavior of the filter factors for large and small γ_i, in particular that both ϕ_i and f_i actually dampen the contributions to $\mathbf{x}_\lambda$ and $\mathbf{x}_{\rm reg}$ corresponding to small γ_i. The only way to obtain useful bounds for $\|L\,(\mathbf{x}_\lambda - \mathbf{x}_{\rm reg})\|$ and $\|A\,(\mathbf{x}_\lambda - \mathbf{x}_{\rm reg})\|$ is to work directly with the vector $(\Phi - F)M_p\Sigma_p^{-1}U_p^T\mathbf{b}$. Our approach to obtaining practical results is to analyze the same model problem as in [20], [21], [48]. We assume that the decay of the Fourier coefficients of the right-hand side, $\beta_i \equiv \mathbf{u}_i^T\mathbf{b}$, is related to the decay of the generalized singular values in the following simple way:

$$\beta_i \equiv \mathbf{u}_i^T\mathbf{b} = \begin{cases} \gamma_i^\alpha, & i = 1, \ldots, p, \\ \gamma_p^\alpha, & i = p+1, \ldots, m. \end{cases} \tag{20}$$

Here $\alpha \ge 0$ is a real parameter that controls the decay of β_i relative to γ_i, and for $\alpha > 1$ the β_i decay to zero *faster* than the γ_i. We shall also assume that the two regularization methods are similar, i.e., $\Phi \approx F$, and that the filter factors f_i corresponding to the largest generalized singular values γ_i are identical to the Tikhonov filter factors ϕ_i. Otherwise, there is no point in comparing $\mathbf{x}_\lambda$ and $\mathbf{x}_{\rm reg}$. For simplicity, we assume that the difference $f_i - \phi_i$ satisfies

$$\begin{aligned} |f_i - \phi_i| &\le c\,\phi_i, & \gamma_i &\le K\,\lambda, \\ f_i &= \phi_i, & \gamma_i &> K\,\lambda, \end{aligned} \tag{21}$$

where c is a small positive constant, and K is a positive constant satisfying $1 \le K < \gamma_p/\lambda$ (thus ensuring that $K\,\lambda < \gamma_p$). Then the norms $\|L\,(\mathbf{x}_\lambda - \mathbf{x}_{\rm reg})\|_\infty$ and $\|A\,(\mathbf{x}_\lambda - \mathbf{x}_{\rm reg})\|_\infty$ can be bounded as follows.

THEOREM 4. *Assume that the Fourier coefficients* $\beta_i = \mathbf{u}_i^T\mathbf{b}$ *are given by* (20), *and that* $\Phi = \mathrm{diag}(\phi_i)$ *and* $F = \mathrm{diag}(f_i)$ *are related by* (21). *Then*

$$\frac{\|L\,(\mathbf{x}_\lambda - \mathbf{x}_{\rm reg})\|_\infty}{\|L\,\mathbf{x}_\lambda\|_\infty} \le \begin{cases} 2\,c, & 0 \le \alpha < 1, \\ 2\,c\,\left(\frac{K\,\lambda}{\gamma_p}\right)^{\alpha-1}, & 1 \le \alpha, \end{cases} \tag{22}$$

[1] The quantity $\|L\,A_\lambda^I - L\,A_k^I\|$ in [21, Thm. 3] is identical to our quantity $\max\{|\phi_i - f_i|/\gamma_i\}$.

$$\frac{\|A(\mathbf{x}_\lambda - \mathbf{x}_{\text{reg}})\|_\infty}{\|A\mathbf{x}_\lambda\|_\infty} \le 2c\left(\frac{K\lambda}{\gamma_p}\right)^\alpha. \tag{23}$$

Proof. The two numerators above are by definition $\|L(\mathbf{x}_\lambda - \mathbf{x}_{\text{reg}})\|_\infty = \|(\Phi - F)M_p\Sigma_p^{-1}U_p^T\mathbf{b}\|_\infty = \max\{|\phi_i - f_i|\,\beta_i/\gamma_i\}$ and $\|A(\mathbf{x}_\lambda - \mathbf{x}_{\text{reg}})\|_\infty = \|(\Phi - F)U_p^T\mathbf{b}\|_\infty = \max\{|\phi_i - f_i|\,\beta_i\}$. If we insert the "model" (20), then for $\gamma_i \le K\lambda$ we obtain

$$|\phi_i - f_i|\,\beta_i/\gamma_i \le c\,\phi_i\,\gamma_i^{\alpha-1} = c\,\frac{\gamma_i^{\alpha+1}}{\gamma_i^2 + \lambda^2}.$$

For $\alpha \ge 1$, we then use $\phi_i \le 1$ and $\gamma_i \le K\lambda$ to obtain the bound $c(K\lambda)^{\alpha-1}$. For $1 \le \alpha < 1$, differentiation of the right-hand expression in the above equation shows that the maximum is attained for $\gamma_i = \lambda$, leading to the upper bound $c\,\frac{1}{2}\lambda^{\alpha-1}$. Regarding $\|L\mathbf{x}_\lambda\|_\infty$, the ∞-norm of a vector is bounded below by the absolute value of any of its elements, and here it is practical to use

$$\|L\mathbf{x}_\lambda\| \ge \begin{cases} \phi_q\gamma_q^{\alpha-1}, & 0 \le \alpha < 1, \\ \phi_p\gamma_p^{\alpha-1}, & 1 \le \alpha, \end{cases}$$

where we have defined the integer q by $\gamma_{q-1} < \lambda \le \gamma_q$. Combining these bounds with $\phi_p^{-1} \le 2$, the bound for $\alpha \ge 1$ in (22) follows. Regarding the bound for $0 \le \alpha < 1$, straightforward analysis shows that $\lambda^{\alpha-1}/(\phi_q\gamma_q^{\alpha-1}) \le 2$ for $\gamma_q \ge \lambda$, and we thus obtain the second bound in (22). The bound in (23) is derived analogously by means of the relations $|f_i - \phi_i|\beta_i \le c\,\phi_i\gamma_i^\alpha \le c(K\lambda)^\alpha$ and $\|A\mathbf{x}_\lambda\|_\infty \ge \phi_p\gamma_p^\alpha$, which hold for all $\alpha \ge 0$. □

Theorem 4 clearly illustrates the importance of the decay of the Fourier coefficients β_i relative to the decay of the generalized singular values γ_i. If β_i decay *more slowly* than γ_i (i.e., if $\alpha < 1$), then (22) gives a large upper bound even if $\phi_i \approx f_i$, i.e., $L\mathbf{x}_{\text{reg}}$ may differ considerably from $L\mathbf{x}_\lambda$. Only if β_i decay somewhat *faster* than γ_i (i.e., if $\alpha > 1$) can we ensure a small upper bound in (22)—provided that the constant c not too large and that $K\lambda$ is somewhat smaller than γ_p. This is, for example, the case when the conjugate gradient method is used as a regularizing iterative scheme, in which case the "exact" filter factors $f_i = \phi_i$ correspond to Ritz values that have converged to eigenvalues of A^TA [24].

Theorem 4 also shows that the residuals $A\mathbf{x}_{\text{reg}} - \mathbf{b}$ and $A\mathbf{x}_\lambda - \mathbf{b}$ will be similar as long as α is somewhat larger than zero, again provided that c is small. That is, the residuals can be similar even if β_i decay slower than γ_i. In other words, similar residuals do not guarantee that $L\mathbf{x}_{\text{reg}} \approx L\mathbf{x}_\lambda$. This fact is in full agreement with the conclusion that we drew from Theorem 3 above.

The requirement that the Fourier coefficients β_i must decay to zero faster than the generalized singular values γ_i is called the *discrete Picard condition* [20], [21], [48], and it is crucial in connection with discrete ill-posed problems. In fact, it is shown in [21, Thm. 2] that this requirement must be satisfied in order to ensure that the Tikhonov regularized solution $\mathbf{x}_\lambda$ will approximate the sought solution, namely, the solution to the "underlying" problem with an unperturbed right-hand side.

In a practical application, the right-hand side $\mathbf{b} = \bar{\mathbf{b}} + \mathbf{e}$ consists of an unperturbed quantity $\bar{\mathbf{b}}$ plus the perturbation $\mathbf{e}$. Assume that $\mathbf{b}$ satisfies the three assumptions from Characterization 2. Then we know that the unperturbed $\bar{\mathbf{b}}$ satisfies the discrete Picard condition, whereas $\mathbf{e}$ typically has almost equal contributions from all left singular vectors $\mathbf{u}_i$, corresponding to $\alpha \approx 0$ in the "model" (20). Hence, the $|\beta_i|$ will typically decay

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to zero for $i = p, p-1, \ldots$ (when $\mathbf{u}_i^T \bar{\mathbf{b}}$ dominates), until $\mathbf{u}_i^T \mathbf{e}$ starts to dominate and the $|\beta_i|$ "level off" at a level determined by the perturbation $\mathbf{e}$. Clearly, the importance of any regularization method with filter factors f_i is to dampen the contributions to the solution, corresponding to the latter β_i, such that the *"regularized" Fourier coefficients* $f_i \beta_i$ satisfy the discrete Picard condition. As long as $\|\mathbf{e}\|$ is somewhat smaller than $\|\bar{\mathbf{b}}\|$ (i.e., there is a satisfactory signal-to-noise ratio in the right-hand side), then both upper bounds in Theorem 4 will be small, and we are thus ensured that $\mathbf{x}_{\mathrm{reg}}$ and $\mathbf{x}_\lambda$ are indeed similar, and $(\|A\,\mathbf{x}_{\mathrm{reg}} - \mathbf{b}\|, \|L\,\mathbf{x}_{\mathrm{reg}}\|)$ will be close to the Tikhonov L-curve.

6. Methods for choosing the regularization parameter. In §3 we mentioned that we would intuitively expect a good regularization parameter λ to produce a regularized solution near the characteristic "corner" of the L-curve because such a λ yields a good balance between a small residual norm $\|A\,\mathbf{x}_\lambda - \mathbf{b}\|$ and a small solution seminorm $\|L\,\mathbf{x}_\lambda\|$. The following observation is important. Notice that the residual vector has the form

$$A\,\mathbf{x}_\lambda - \mathbf{b} = (A\,\bar{\mathbf{x}}_0 - \mathbf{b}) - A(\bar{\mathbf{x}}_0 - \bar{\mathbf{x}}_\lambda) - A(\bar{\mathbf{x}}_\lambda - \mathbf{x}_\lambda), \tag{24}$$

where $\bar{\mathbf{x}}_0$ is the exact unregularized solution to the unperturbed problem, $\bar{\mathbf{x}}_0 - \bar{\mathbf{x}}_\lambda$ is the regularization error, and $\bar{\mathbf{x}}_\lambda - \mathbf{x}_\lambda$ is the perturbation error. Equation (24) shows that a large regularization error also means a large residual vector. Moreover, we know from the analysis in §3 that a large perturbation error implies a large seminorm $\|L\,\mathbf{x}_\lambda\|$. This means that a solution near the L-curve's "corner," in addition to balancing the residual norm and the solution seminorm, also tends to balance the regularization and perturbation errors. This is yet another reason for choosing a regularization parameter that gives a solution near the "corner" of the L-curve.

We now show that different methods for choosing λ are actually related to locating this "corner." We focus our attention on Tikhonov regularization, knowing that the results carry over to methods that are similar to it.

6.1. The discrepancy principle. One method that has attained a widespread interest is the *discrepancy principle*, usually attributed to Morozov [31]. If the ill-posed problem is consistent, i.e., if $\delta_0 = 0$, and if only the right-hand side is perturbed, then the idea is simply to select the regularization parameter λ so that the residual norm is equal to an a priori upper bound $\delta_{\mathbf{e}}$ for the norm of the errors $\mathbf{e}$ in the right-hand side, i.e.,

$$\|A\,\mathbf{x}_\lambda - \mathbf{b}\| = \delta_{\mathbf{e}}, \quad \text{where } \|\mathbf{e}\| \le \delta_{\mathbf{e}}. \tag{25}$$

If we assume that the ill-posed problem satisfies the assumptions in Characterization 2, then the expected value of $\|\mathbf{e}\|$ is $\sqrt{m}\,\sigma_0$. Equation (25), therefore, corresponds to choosing a solution that appears on the L-curve a little to the right of the "corner," which, according to Characterization 2, is approximately at $(\sigma_0\sqrt{m-n+p}, \|L\,\bar{\mathbf{x}}_0\|)$. Notice that if $\|\mathbf{e}\|$ is not known a priori, and if $\mathbf{e}$ has zero mean and covariance matrix $\sigma_0^2 I_m$ (i.e., it satisfies the second assumption in Characterization 2), then σ_0 can be estimated by monitoring the function $\mathcal{V}(\lambda)$ [47, p. 68] defined by

$$\mathcal{V}(\lambda) \equiv \frac{\|A\,\mathbf{x}_\lambda - \mathbf{b}\|^2}{\mathcal{T}(\lambda)}. \tag{26}$$

Here, for convenience, we have defined another function $\mathcal{T}(\lambda)$ (which can be considered as the "degrees of freedom" [47, pp. 63, 68]):

$$\mathcal{T}(\lambda) \equiv \mathrm{trace}(I_m - A(A^TA + \lambda^2 L^TL)^{-1}A^T) = m - n + \sum_{i=1}^{p} \frac{\lambda^2}{\gamma_i^2 + \lambda^2}. \tag{27}$$

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If $\mathcal{V}$ is plotted versus λ^{-1}, then on a broad scale the graph of $\mathcal{V}$ first decreases, then "levels off" at a plateau that is the estimate of σ_0^2, and eventually decreases to zero for small λ. The estimate of $\|\mathbf{e}\|^2$ is equal to m times the value of $\mathcal{V}$ at the plateau.

The *generalized discrepancy principle* [31, p. 53] also takes into account errors E in the matrix A, as well as the incompatibility measure δ_0 in (10). Let $\delta_{\mathbf{e}}$ and δ_E denote upper bounds for $\|\mathbf{e}\|$ and $\|E\|$, respectively. Then the generalized discrepancy principle amounts to choosing λ such that[2]

$$\|A\mathbf{x}_\lambda - \mathbf{b}\| = \delta_0 + \delta_{\mathbf{e}} + \delta_E \|\mathbf{x}_0\|, \tag{28}$$

where $\|\mathbf{x}_0\|$ is the norm of the unregularized solution $\mathbf{x}_0$ (9). If the user has an a priori upper bound for $\|\mathbf{x}_\lambda\|$, the norm of the desired solution, then this upper bound should be substituted for $\|\mathbf{x}_0\|$ in (28). Estimates for $\|E\|$, based solely on statistical information about the errors E, can be found in [8], [16]. An alternative formulation to (28) is [31, p. 58]:

$$\|A\mathbf{x}_\lambda - \mathbf{b}\| = \delta_0 + \delta_{\mathbf{e}} + \Delta_{E,L} \|L\mathbf{x}_\lambda\|, \tag{29}$$

where $\Delta_{E,L}$ is an upper bound for $\max_{L\mathbf{x}\neq\mathbf{0}}\{\|E\mathbf{x}\|/\|L\mathbf{x}\|\}$, the largest generalized singular value of the pair (E, L). In particular, if $L = I_n$, then $\Delta_{E,L} = \delta_E$. The regularized solution computed by means of (29) corresponds to that point in the $\|A\mathbf{x}_\lambda - \mathbf{b}\|$–$\|L\mathbf{x}_\lambda\|$ plane where the line (29) intersects the L-curve. The approach in (29) is appealing because it does not involve an a priori upper bound for $\|\mathbf{x}_\lambda\|$.

All three formulations (25), (28), (29) of the discrepancy principle are based on a conservative choice of the residual norm $\|A\mathbf{x}_\lambda - \mathbf{b}\|$. In terms of the L-curve, they all produce regularized solutions appearing to the right of the "corner," and this is particularly pronounced if $\delta_E \neq 0$. Hence the claim in [25, p. 96] that "the discrepancy principle oversmooths the real solution." Wahba [47, p. 63] has come to the same conclusion from a statistical viewpoint.

6.2. The quasi-optimality criterion. The second method that we shall consider here is the *quasi-optimality criterion*; cf., e.g., [31, §27], which amounts to finding the regularization parameter λ that minimizes the function

$$Q(\lambda) \equiv \left\| \lambda^2 \frac{d\mathbf{x}_\lambda}{d(\lambda^2)} \right\| = \frac{1}{2} \left\| \lambda \frac{d\mathbf{x}_\lambda}{d\lambda} \right\|. \tag{30}$$

We note in passing that the second step of iterated Tikhonov regularization [31, p. 238] leads to the solution $(A^TA + \lambda^2 L^TL)^{-1}(A^T\mathbf{b} + \lambda^2 L^TL\mathbf{x}_\lambda) = \mathbf{x}_\lambda - \lambda^2 d\mathbf{x}_\lambda/d(\lambda^2)$, such that minimization of $Q(\lambda)$ minimizes the correction to $\mathbf{x}_\lambda$ in this solution. Morozov [31, p. 240], regarding the quasi-optimality criterion writes: "Unfortunately, it has not been possible to justify this technique for choosing the parameter although it is widely used for solving unstable problems." Recently, by studying a standard-form model problem satisfying the discrete Picard condition, Kitagawa [26] demonstrated that the λ which minimizes $Q(\lambda)$ seeks to minimize the error $\bar{\mathbf{x}}_0 - \mathbf{x}_\lambda$ in the solution. We shall here give a related, but somewhat more heuristic, analysis that relates the minimization of $Q(\lambda)$ to the "corner" of the L-curve.

If we insert the expression (6) for $\mathbf{x}_\lambda$ into (30) and make use of the behavior of the filter factors (i.e., $\phi_i \approx 1$ for large γ_i and $\phi_i \approx 0$ for small γ_i), then we obtain

$$Q(\lambda)^2 = \sum_{i=1}^{p} \phi_i^2 (1-\phi_i)^2 \left(\frac{\beta_i}{\sigma_i}\right)^2 \approx \sum_{\gamma_i \geq \lambda} (1-\phi_i)^2 \left(\frac{\beta_i}{\sigma_i}\right)^2 + \sum_{\gamma_i < \lambda} \phi_i^2 \left(\frac{\beta_i}{\sigma_i}\right)^2.$$

[2]The sharper right-hand side $(\delta_0^2 + (\delta_e + \delta_E\|\mathbf{x}_0\|)^2)^{1/2}$ also appears in the literature.

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Now let $\bar{\beta}_i \equiv \mathbf{u}_i^T \bar{\mathbf{b}}$ denote the Fourier coefficients of the unperturbed right-hand side $\bar{\mathbf{b}}$, and assume that $\bar{\mathbf{b}}$ satisfies the discrete Picard condition and that λ is chosen so as to produce a solution near the L-curve's "corner." Then we have $\bar{\beta}_i/\sigma_i \approx 0$ for small σ_i and small γ_i, while $\bar{\beta}_i \approx \beta_i$ for large σ_i and large γ_i. Using these approximations, we obtain the following approximate expression for the regularization and perturbation errors:

$$\|L\,(\bar{\mathbf{x}}_0 - \bar{\mathbf{x}}_\lambda)\|^2 = \sum_{i=1}^{p} (1-\phi_i)^2 \left(\frac{\bar{\beta}_i}{\sigma_i}\right)^2 \approx \sum_{\gamma_i \geq \lambda} (1-\phi_i)^2 \left(\frac{\beta_i}{\sigma_i}\right)^2,$$

$$\|L\,(\bar{\mathbf{x}}_\lambda - \mathbf{x}_\lambda)\|^2 = \sum_{i=1}^{p} \phi_i^2 \left(\frac{\bar{\beta}_i - \beta_i}{\sigma_i}\right)^2 \approx \sum_{\gamma_i > \lambda} \phi_i^2 \left(\frac{\beta_i}{\sigma_i}\right)^2.$$

Thus, we have obtained the following approximate expression for the quasi-optimality function:

$$Q(\lambda)^2 \approx \|L(\bar{\mathbf{x}}_0 - \bar{\mathbf{x}}_\lambda)\|^2 + \|L(\bar{\mathbf{x}}_\lambda - \mathbf{x}_\lambda)\|^2. \tag{31}$$

In other words, the minimizer of $Q(\lambda)$ seeks to find a good compromise between minimization of the regularization error $\bar{\mathbf{x}}_0 - \bar{\mathbf{x}}_\lambda$ and the perturbation error $\bar{\mathbf{x}}_\lambda - \mathbf{x}_\lambda$. And according to the discussion in the beginning of this section, this criterion is exactly the same as localizing the "corner" of the L-curve.

6.3. Generalized cross-validation. Another popular method for choosing the regularization parameter λ is the *generalized cross-validation* (GCV) method due to Golub, Heath, and Wahba [12]. GCV is based on statistical considerations, namely, that a good value of the regularization parameter should predict missing data values. In this way no a priori knowledge about the error norms is required. GCV leads to choosing λ as the minimizer of the GCV function $\mathcal{G}(\lambda)$, defined by

$$\mathcal{G}(\lambda) \equiv \frac{\|A\,\mathbf{x}_\lambda - \mathbf{b}\|^2}{(\mathcal{T}(\lambda))^2} = \frac{\mathcal{V}(\lambda)}{\mathcal{T}(\lambda)}. \tag{32}$$

As already mentioned above, if the errors $\mathbf{e}$ in the right-hand side satisfy the second assumption in Characterization 2, then the general behavior of $\mathcal{V}$ as a function of λ^{-1} is to decrease until it "levels off" at a plateau approximately at σ_0^2. The transition between $\mathcal{V}$ being a decreasing function and a function that "levels off" takes place in a (usually small) λ-interval, and obviously it is for the same λ-interval that the L-curve has its characteristic "corner." The GCV method seeks to locate this "corner" implicitly by instead locating the transition of the function $\mathcal{V}$. But instead of working with $\mathcal{V}$, GCV uses the function $\mathcal{G}$. Note that the denominator $\mathcal{T}$, given by (27), is a monotonically increasing function of λ, such that $\mathcal{G}$ given by (32) has a *minimum* in the above-mentioned transition interval. Hence, GCV replaces the problem of locating the transition point for $\mathcal{V}$ by a numerically well-defined problem, namely, that of finding the minimum for the GCV function $\mathcal{G}$. Figure 3 in §7 shows a typical example of a GCV function.

Apart from the fact that $\mathcal{T}(\lambda)$ is an increasing function, from $m-n$ for $\lambda = 0$ to $m-n+p$ for $\lambda \to \infty$, it is difficult to characterize this function further. On the other hand, the above argument that GCV locates the "corner" of the L-curve relies on $\mathcal{T}(\lambda)$ increasing slowly throughout the interval $\gamma_1 \leq \lambda \leq \gamma_p$. The following theorem sheds more light on this aspect.

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THEOREM 5. *If there is a constant ratio c between all the generalized singular values, such that $\gamma_i = c\gamma_{i+1}$ (with $0 < c < 1$), then*

$$m - n + k_\lambda - \frac{1}{1-c^2} \le \mathcal{T}(\lambda) \le m - n + k_\lambda + \frac{1}{1-c^2}, \tag{33}$$

where k_λ is the number of γ_i less than λ, i.e., $\gamma_{k_\lambda} < \lambda \le \gamma_{k_\lambda+1}$.

Proof. To derive (33) from (27), we must consider the quantity

$$\sum_{i=1}^{p}(1-\phi_i) = \sum_{i=1}^{k_\lambda}\left(1-\left(1+\left(\frac{\lambda}{\gamma_i}\right)^2\right)^{-1}\right) + \sum_{i=k_\lambda+1}^{p}\left(1+\left(\frac{\gamma_i}{\lambda}\right)^2\right)^{-1}$$

with k_λ defined above. It is easy to show that

$$\sum_{i=1}^{k_\lambda}\left(1-\left(1+\left(\frac{\lambda}{\gamma_i}\right)^2\right)^{-1}\right) = k_\lambda - \sum_{i=1}^{k_\lambda}\left(1+\left(\frac{\lambda}{\gamma_i}\right)^2\right)^{-1},$$

$$0 \le \sum_{i=1}^{k_\lambda}\left(1+\left(\frac{\lambda}{\gamma_i}\right)^2\right)^{-1} \le \sum_{i=1}^{k_\lambda}\left(\frac{\gamma_i}{\lambda}\right)^2 = \left(c^{2(k_\lambda-1)} + \cdots + c^2 + 1\right)\left(\frac{\gamma_{k_\lambda}}{\lambda}\right)^2,$$

$$0 \le \sum_{i=k_\lambda+1}^{p}\left(1+\left(\frac{\gamma_i}{\lambda}\right)^2\right)^{-1} \le \sum_{i=k_\lambda+1}^{p}\left(\frac{\lambda}{\gamma_i}\right)^2 = \left(\frac{\lambda}{\gamma_{k_\lambda+1}}\right)^2\left(1 + c^2 + \cdots + c^{2(p-k_\lambda-1)}\right).$$

Using these relations together with $\gamma_{k_\lambda} < \lambda \le \gamma_{k_\lambda+1}$ and

$$1 + c^2 + \cdots + c^{2(q-1)} = \frac{1-c^{2q}}{1-c^2} \le \frac{1}{1-c^2},$$

we arrive at (33). □

Theorem 5 shows that for this particular geometric distribution of generalized singular values—which resembles many practical applications—the variation of $\mathcal{T}$ with λ indeed takes place throughout the interval $[\gamma_1, \gamma_p]$, so that the function $\mathcal{G}$ defined in (32) has a minimum that corresponds to the "corner" of the L-curve.

The GCV method has proven its usefulness in numerous applications [47]. However, two difficulties are associated with this method: the minimum of the GCV function is often very flat and, therefore, difficult to locate numerically [44], and the method may fail to compute the correct λ when the errors are highly correlated [47, p. 65]. In the latter case, the graph of $\mathcal{V}$ may not have a plateau, in which case the GCV function $\mathcal{G}$ does not attain its minimum for a λ corresponding to the L-curve's "corner." We illustrate this difficulty by a numerical example in the next section, in particular, in Fig. 5.

6.4. The L-curve criterion. Inspired by the observations and characterization of the L-curve in the present paper, Hansen and O'Leary proposed a new method for choosing the regularization parameter λ, based on an algorithm that locates the "corner" of the L-curve [23]. They define the "corner" as the point on the L-curve with maximum curvature, and they give an algorithm for computing this "corner." They also extend the ideas from the L-curve for Tikhonov regularization to other regularization methods, including those with a discrete regularization parameter (such as truncated SVD). As we shall illustrate in §7, this new L-curve criterion for chosing λ is often more robust to correlated errors than the GCV method.

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6.5. A perturbation bound. We conclude this section with an unusual perturbation bound for the solution that directly involves the shape of the L-curve. The perturbation bound, given in Corollary 7 below, is based on the following theorem from [38], [39].

THEOREM 6. *Let $\mathcal{F}$ denote the function that maps $\|A\mathbf{x}_\lambda - \mathbf{b}\|$ to $\|L\mathbf{x}_\lambda\|$:*

$$\|L\mathbf{x}_\lambda\| = \mathcal{F}(\|A\mathbf{x}_\lambda - \mathbf{b}\|). \tag{34}$$

Also, let $\mathbf{e}$ and E denote the perturbations in $\mathbf{b}$ and A, respectively, and let $\bar{\mathbf{x}}_{\bar{\lambda}}$ denote the unperturbed solution for which

$$\|(A - E)\,\bar{\mathbf{x}}_{\bar{\lambda}} - (\mathbf{b} - \mathbf{e})\| = \|A\mathbf{x}_\lambda - \mathbf{b}\|. \tag{35}$$

Then

$$\|L\,(\mathbf{x}_\lambda - \bar{\mathbf{x}}_{\bar{\lambda}})\| < 2\sqrt{(\mathcal{F}(\delta - \delta_e))^2 - (\mathcal{F}(\delta + \delta_e))^2}, \tag{36}$$

where $\delta = \|A\mathbf{x}_\lambda - \mathbf{b}\|$, $\delta_e = \|\mathbf{e}\| + \eta_e\,\|E\|$, and η_e is the solution to $\eta_e = \mathcal{F}(\delta + \delta_e)$.

COROLLARY 7. *With the same notation as in Theorem 6, we have*

$$\frac{\|L\,(\mathbf{x}_\lambda - \bar{\mathbf{x}}_{\bar{\lambda}})\|}{\|\mathbf{x}_\lambda\|} < 4\sqrt{|\mathcal{F}'(\delta)|\,\|A\|\,\epsilon}\,(1 + O(\epsilon))\,, \tag{37}$$

with

$$\epsilon = \frac{\|\mathbf{e}\|}{\|\mathbf{b}_\lambda\|} + \frac{\|E\|}{\|A\|},$$

where $\mathbf{b}_\lambda = A\mathbf{x}_\lambda$, and $\mathcal{F}'(\delta)$ denotes the derivative of $\mathcal{F}$ at $\delta = \|A\mathbf{x}_\lambda - \mathbf{b}\|$.

Proof. By means of the Taylor expansion $\mathcal{F}(\delta \pm \delta_e) = \mathcal{F}(\delta) \pm \delta_e\mathcal{F}'(\delta) + O(\delta_e^2)$, we readily obtain

$$(\mathcal{F}(\delta - \delta_e))^2 - (\mathcal{F}(\delta + \delta_e))^2 = -4\,\mathcal{F}(\delta)\,\delta_e\,\mathcal{F}'(\delta) + O(\delta_e^2).$$

Since $\mathcal{F}$ is a monotonically decreasing function with $-\mathcal{F}'(\delta) = |\mathcal{F}'(\delta)|$, we know that $\mathcal{F}(\delta) \geq \mathcal{F}(\delta + \delta_e)$, and we can insert the upper bound $\mathcal{F}(\delta) = \|L\mathbf{x}_\lambda\|$ for η_e. These relations, together with $\|\mathbf{b}_\lambda\| \leq \|A\|\,\|\mathbf{x}_\lambda\|$, yield (37). □

To guarantee a small perturbation bound in (37), we must choose the regularization parameter λ so that $|\mathcal{F}'(\|L\mathbf{x}_\lambda\|)|$ is small, i.e., $\mathbf{x}_\lambda$ should correspond to a solution on that part of the L-curve immediately to the right of the "corner." This principle, combined with one or more of the above-mentioned methods and (if possible) with a visual inspection of the L-curve, should lead to a good choice of the regularization parameter in most cases.

7. Numerical examples. The purpose of this last section is to illustrate some of the topics discussed in the previous sections, and in particular we will focus on the behavior of the L-curve and the GCV function for two different perturbations: white noise and correlated noise.

Throughout this section, we consider a discrete ill-posed problem, which is a discretization of a Fredholm integral equations of the first kind:

$$\int_a^b K(s,x)\,f(x)\,dx = g(s), \qquad c \leq s \leq d, \tag{38}$$

where K is the kernel, g is the right-hand side, and f is the unknown solution.

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The particular integral equation that we shall use is a one-dimensional model problem in image reconstruction from [36, §5], where an image is blurred by a known point-spread function. The desired solution f is given by

$$f(x) = 2\,\exp(-4(x-0.5)^2) + \exp(-4(x+0.5)^2), \qquad -\frac{\pi}{2} \le x \le \frac{\pi}{2}, \tag{39}$$

while the kernel K is the point spread function of an infinitely long slit, given by

$$K(s,x) = \left((\cos s + \cos x)\frac{\sin \psi_{s,x}}{\psi_{s,x}}\right)^2, \quad -\frac{\pi}{2} \le s \le \frac{\pi}{2}, \tag{40}$$

with the function $\psi_{s,x}$ given by

$$\psi_{s,x} = \pi\,(\sin s + \sin x).$$

We use simple collocation with n equidistantly spaced points in $[-\pi/2, \pi/2]$ to derive the matrix A and the exact solution $\bar{\mathbf{x}}_0$. Then we compute the exact right-hand side as $\bar{\mathbf{b}} = A\,\bar{\mathbf{x}}_0$. The order of the matrix is $n = 64$ in all our examples.

7.1. Perturbations by white noise. First we consider problems where the right-hand side is perturbed by uncorrelated errors (white noise), i.e., the elements e_i of the perturbation $\mathbf{e}$ are normally distributed with zero mean and standard deviation 10^{-4}. Here, we choose a matrix L equal to the second derivative operator, $L = \text{tridiag}(1, -2, 1)$, with $p = n - 2$ rows.

Figure 1 shows the Tikhonov L-curve for this problem, together with the points corresponding to the truncated GSVD solutions (7). This example illustrates that truncated GSVD solutions can be similar to regularized solutions computed by Tikhonov regularization. Notice the distinct "corner" that clearly shows the value of the regularization parameter where the perturbation $\mathbf{e}$ starts to dominate the solution. Three values of λ corresponding to solutions near the "corner" are indicated, and these particular solutions are shown in Fig. 2, together with the exact solution $\bar{\mathbf{x}}_0$. The solution computed with $\lambda = 2{\cdot}10^{-3}$ lies closest to the L-curve's "corner" and is also the best judging from Fig. 2. The solution computed with $\lambda = 2{\cdot}10^{-2}$ is oversmoothed; i.e., it is too rigid to follow the variations in the desired solution; and the solution computed with $\lambda = 2{\cdot}10^{-4}$ is clearly undersmoothed and has a too large component from the perturbation $\mathbf{e}$.

The GCV function $\mathcal{G}(\lambda)$ for this problem is shown in Fig. 3, and it has the typical appearance of GCV functions: there is a very flat part and a steeper part of the curve. The minimum of $\mathcal{G}(\lambda)$ occurs approximately at $\lambda = 5{\cdot}10^{-3}$, which definitely is very near the "corner" of the L-curve. Thus, for this problem the GCV method indeed determines a good regularization parameter.

7.2. Perturbations by correlated noise. Next, we illustrate the behavior of the L-curve and the GCV function when the errors are highly correlated. For simplicity, we here take $L = I$, the identity matrix. Two types of correlated errors are considered:

1. Filtered white noise $\mathbf{e}$, generated with the formula $e_i = a\,e_{i-1} + \epsilon_i$, where $0 \le a \le 1$ and ϵ_i is white noise;
2. Errors from a regular smoothing of the matrix A and the right-hand side $\bar{\mathbf{b}}$:

$$\tilde{a}_{ij} = a_{ij} + \mu\,(a_{i-1,j} + a_{i+1,j} + a_{i,j-1} + a_{i,j+1}),$$

$$b_i = \bar{b}_i + \mu\,(\bar{b}_{i-1} + \bar{b}_{i+1}).$$

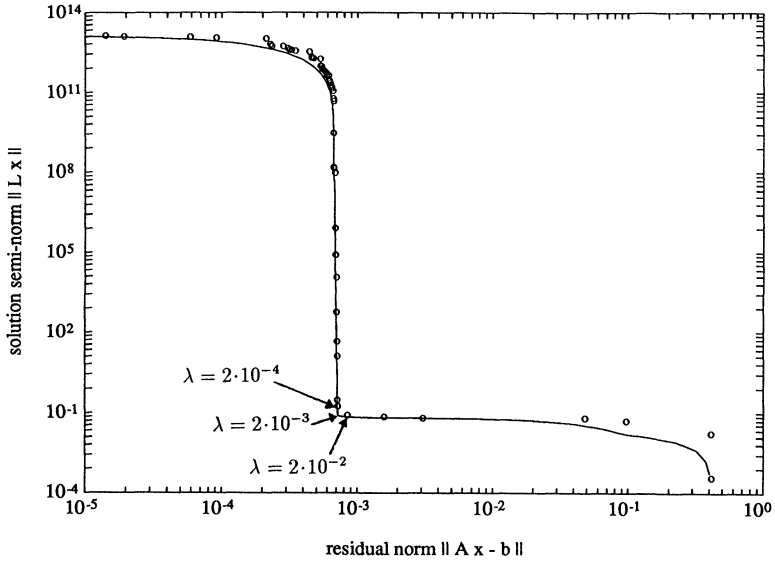


FIG. 1. *The Tikhonov L-curve* $(\|A\mathbf{x}_\lambda - \mathbf{b}\|, \|L\mathbf{x}_\lambda\|)$ *for an example with uncorrelated errors (white noise). Also shown as circles are the truncated* GSVD *solutions.*

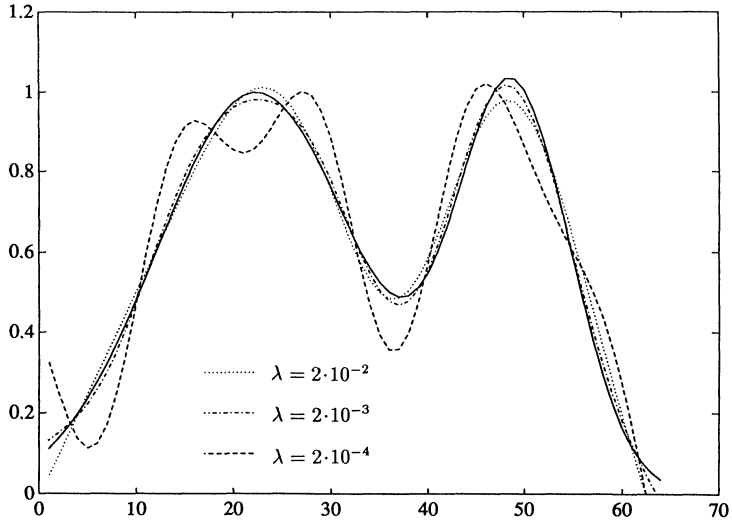


FIG. 2. *The exact solution* $\bar{\mathbf{x}}_0$ *(solid line) and three regularized solutions* $\mathbf{x}_\lambda$ *corresponding to the three values of* λ *shown in Fig.* 1.

Hence, for the right-hand side, the perturbation $\mathbf{e}$ has elements $e_i = b_i - \bar{b}_i$; and similarly for the matrix.

Both types of errors give rise to Fourier coefficients $\mathbf{u}_i^T\mathbf{e}$ (where $\mathbf{u}_i$ are the left singular vectors of $\tilde{A}$) that decay with increasing i; i.e., $\mathbf{e}$ has more low-frequency than high-frequency components. The "spectrum" $|\mathbf{u}_i^T\mathbf{e}|$ for type-1 errors is much flatter than that for type-2 errors.

Regarding the errors of type 1, both the L-curve and the GCV function behave precisely as in the case of white noise. Hence, both the L-curve criterion and the GCV method compute good regularization parameters. No results are shown for this case.

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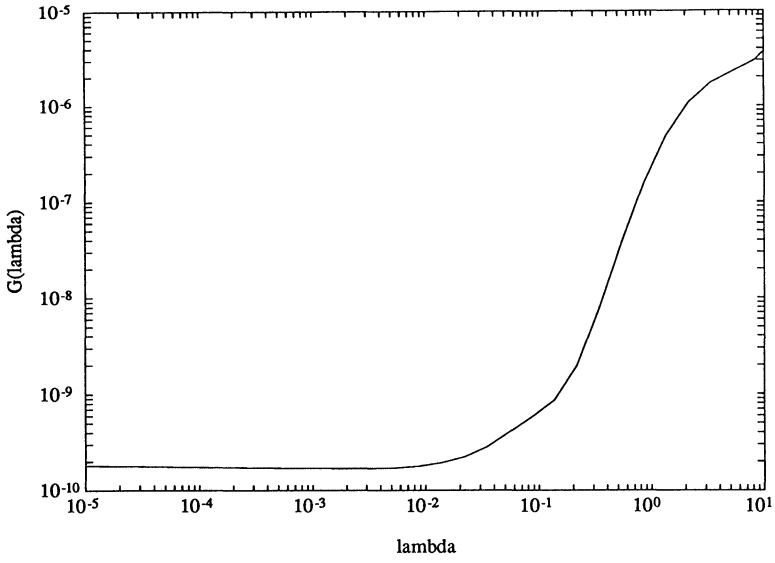


FIG. 3. *The* GCV *function* $\mathcal{G}(\lambda)$ *for the same example as in Figs.* 1 *and* 2. *The minimum of* $\mathcal{G}(\lambda)$ *is attained for* $\lambda \approx 5 \cdot 10^{-3}$.

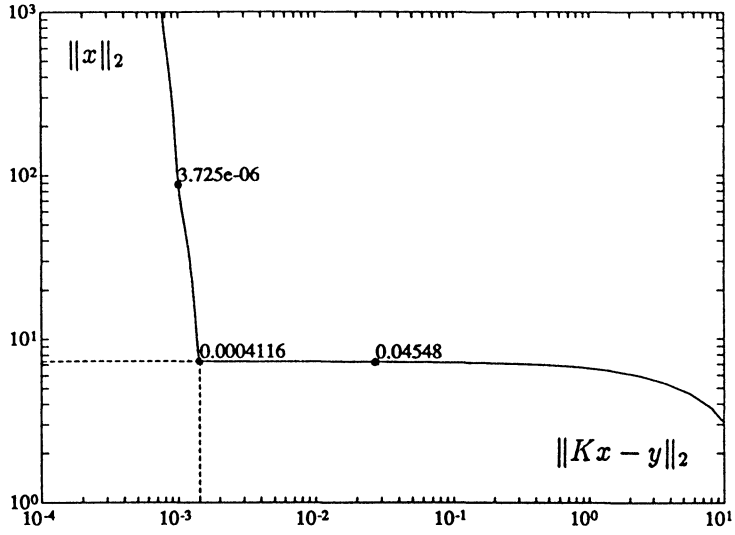


FIG. 4. *The* L-*curve for a problem with highly correlated errors. The "corner" is still a distinct feature of this* L-*curve.*

Regarding the errors of type 2, the situation is quite different. These errors may represent sampling errors, because some averaging of the signal always occurs during the sampling of data. They may also represent the approximation errors involved in computing A and $\mathbf{b}$ by means of a Galerkin-type method, say, where some "local" integration is performed.

Figure 4 shows the L-curve for an example with smoothing parameter $\mu = 0.05$. The three dots on the L-curve correspond to regularization parameters given by the associated numbers. There is a distinct "corner" on the L-curve for $\lambda \approx 4 \cdot 10^{-4}$, and the corresponding regularized solution $\mathbf{x}_\lambda$ is a good approximation to the exact solution $\bar{\mathbf{x}}_0$,

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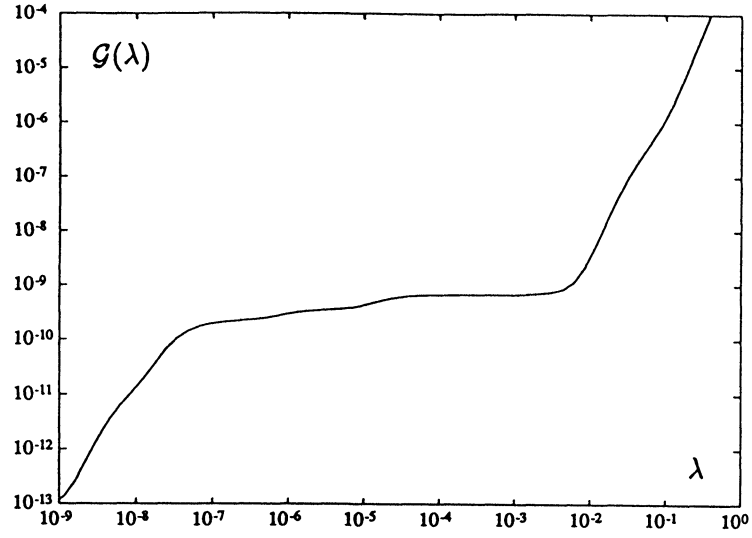


FIG. 5. *The* GCV *function* $\mathcal{G}(\lambda)$ *for the same problem as in Fig.* 4. *The* GCV *function attains its minimum for* λ *of the order of the machine precision (outside the plot).*

the relative error being $\|\mathbf{x}_\lambda - \bar{\mathbf{x}}_0\|_2 / \|\bar{\mathbf{x}}_0\|_2 = 0.12$. The GCV function $\mathcal{G}(\lambda)$ for the same problem is shown in Fig. 5. The maximum is attained for λ of the order of the machine precision (located outside of the plot). In this situation, the GCV method completely fails to compute a useful solution, and the relative error in $\mathbf{x}_\lambda$ is $6.7 \cdot 10^5$.

Apparently, GCV "mistakes" the correlated errors for being part of the wanted signal, and thus chooses a very small regularization parameter that only filters out the white noise in $\mathbf{b}$ due to the rounding errors. The L-curve, on the other hand, leads to a regularization parameter that indeed filters out the correlated errors because they represent a signal that does not satisfy the discrete Picard condition (assumption 1 in Characterization 2); i.e., the coefficients $\mathbf{u}_i^T \mathbf{e}$ do not decay as fast as the singular values.

The essential difference between the GCV method and the L-curve criterion is that the L-curve criterion is able to recognize correlated errors as long as they do not satisfy the discrete Picard condition, while the GCV method may fail to do so. This is essentially because the L-curve criterion combines information about the residual norm with information about the solution (semi)norm, whereas the GCV method only uses the information about the residual norm. For more details about these aspects, see [23].

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